$$\cos\theta = \sqrt{1-\sin^2\theta}$$

$$= \sqrt{1-\frac{25}{169}}$$

$$= \frac{12}{13} \qquad \ldots (2)$$

(1) மற்றும் (2) லிருந்து,

$$\therefore \cos\left(\sin^{-1}\frac{5}{13}\right) = \cos\theta = \frac{12}{13}$$

(ii) $\left(\cos^{-1}\frac{8}{17}\right) = \theta$ என்க

$$\cos\theta = \frac{8}{17}$$

$$\sin\theta = \sqrt{1-\cos^2\theta}$$

$$= \sqrt{1-\frac{64}{289}}$$

$$= \frac{15}{17} \qquad \ldots (2)$$

(1) மற்றும் (2) லிருந்து,

$$\tan\left(\cos^{-1}\frac{8}{17}\right) = \tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$= \frac{\frac{15}{17}}{\frac{8}{17}} = \frac{15}{8}$$

எடுத்துக்காட்டு 4.25

நிறுவுக:

(i) $\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)=\tan^{-1}\left(\frac{2}{9}\right)$

(ii) $\cos^{-1}\left(\frac{4}{5}\right)+\tan^{-1}\left(\frac{3}{5}\right)=\tan^{-1}\left(\frac{27}{11}\right)$

தீர்வு

(i) $\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)$

$$= \tan^{-1}\left[\frac{\frac{1}{7}+\frac{1}{13}}{1-\left(\frac{1}{7}\right)\left(\frac{1}{13}\right)}\right]$$

$$= \tan^{-1}\left(\frac{20}{90}\right)$$

$$= \tan^{-1}\left(\frac{2}{9}\right)$$

(ii) $\cos^{-1}\left(\frac{4}{5}\right) = \theta$ என்க.

எனவே, $\cos\theta = \frac{4}{5} \Rightarrow \sin\theta = \frac{3}{5}$

$$\therefore \tan\theta = \frac{3}{4} \Rightarrow \theta = \tan^{-1}\left(\frac{3}{4}\right)$$

$$\therefore \cos^{-1}\left(\frac{4}{5}\right) = \tan^{-1}\left(\frac{3}{4}\right)$$

இப்பொழுது $\cos^{-1}\left(\frac{4}{5}\right)+\tan^{-1}\left(\frac{3}{5}\right)$

$$= \tan^{-1}\left(\frac{3}{4}\right)+\tan^{-1}\left(\frac{3}{5}\right)$$

$$= \tan^{-1}\left(\frac{\frac{3}{4}+\frac{3}{5}}{1-\frac{3}{4}\times\frac{3}{5}}\right)$$

$$= \tan^{-1}\left(\frac{27}{11}\right)$$

எடுத்துக்காட்டு 4.26

மதிப்பிடுக : $\tan\left[\frac{\pi}{4}-\tan^{-1}\left(\frac{1}{8}\right)\right]$

தீர்வு

$\tan^{-1}\left(\frac{1}{8}\right) = \theta$ என்க

எனவே $\tan\theta = \frac{1}{8}$

$$\therefore \tan\left[\frac{\pi}{4}-\tan^{-1}\left(\frac{1}{8}\right)\right] = \tan\left(\frac{\pi}{4}-\theta\right)$$

$$= \frac{\tan\frac{\pi}{4}-\tan\theta}{1+\tan\frac{\pi}{4}\tan\theta}$$

$$= \frac{1-\frac{1}{8}}{1+\frac{1}{8}} = \frac{7}{9}$$

எடுத்துக்காட்டு 4.27

தீர்க்க : $\tan^{-1}\left(\frac{x-1}{x-2}\right)+\tan^{-1}\left(\frac{x+1}{x+2}\right)=\frac{\pi}{4}$

தீர்வு:

$$\tan^{-1}\left(\frac{x-1}{x-2}\right)+\tan^{-1}\left(\frac{x+1}{x+2}\right)$$

$$= \tan^{-1}\left[\frac{\frac{x-1}{x-2}+\frac{x+1}{x+2}}{1-\left(\frac{x-1}{x-2}\right)\left(\frac{x+1}{x+2}\right)}\right] = \tan^{-1}\left(\frac{2x^2-4}{-3}\right)$$

$\tan^{-1}\left(\frac{x-1}{x-2}\right)+\tan^{-1}\left(\frac{x+1}{x+2}\right)=\frac{\pi}{4}$ என

கொடுக்கப்பட்டுள்ளது.

$$\therefore \tan^{-1}\left(\frac{2x^2-4}{-3}\right) = \frac{\pi}{4}$$

$$\frac{2x^2-4}{-3} = \tan\frac{\pi}{4} = 1$$

$$2x^2-4 = -3$$

$$\Rightarrow 2x^2-1 = 0$$

$$\Rightarrow x = \pm\frac{1}{\sqrt{2}}$$

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